

SOUTHAMPTON, United KingdomWMO: **038650**Lat: **50.950N**Lon: **1.357W**Elev: **13**StdP: **101.16**Time Zone: **0.00 (EUW)**Period: **90-01**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.2	(c) -0.8	(d) -6.6	(e) 2.2	(f) 1.8	(g) -4.9	(h) 2.5	(i) 2.5	(j) 10.9	(k) 9.3	(l) 9.4	(m) 9.4	(n) 2.1	(o) 50	(p) 0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.0	(c) 26.2	(d) 18.3	(e) 24.5	(f) 17.4	(g) 23.0	(h) 16.9	(i) 19.4	(j) 24.1	(k) 18.5	(l) 22.8	(m) 17.7	(n) 21.5	(o) 2.8	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.7	20.9	16.9	12.1	20.0	16.2	11.5	19.4	55.1	24.1	52.3	22.8	49.9	21.5	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.2	7.1	6.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-4.7	29.2	1.7	2.0	-5.9	30.7	-6.9	31.8	-7.8	32.9	-9.0	34.3		
			-5.4	21.8	1.8	2.0	-6.6	23.3	-7.7	24.5	-8.7	25.6	-10.0	27.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	6.2	6.4	8.2	9.6	13.4	15.5	18.0	18.2	15.3	11.9	8.7	6.3		
	DBStd	5.11	3.03	3.06	2.42	2.55	3.04	2.31	2.16	2.39	2.19	2.79	2.99	3.36		
	HDD10.0	524	121	102	64	38	6	0	0	0	0	13	59	121		
	HDD18.3	2553	377	334	313	263	156	91	33	31	94	198	291	373		
	CDD10.0	1073	2	2	9	26	109	165	247	255	158	74	19	6		
	CDD18.3	61	0	0	0	0	2	5	22	28	3	1	0	0		
	CDH23.3	286	0	0	0	0	18	29	104	126	8	1	0	0		
	CDH26.7	39	0	0	0	0	1	3	11	24	0	0	0	0		
(14) Wind	WSAvg	2.6	3.0	3.0	2.7	2.9	2.5	2.4	2.2	2.1	2.2	2.5	2.4	2.9		
(15) Precipitation	PrecAvg	815	94	66	51	55	49	54	47	56	59	93	101	89		
	PrecMax	1123	235	168	135	149	105	146	118	135	140	167	216	180		
	PrecMin	621	17	6	4	0	8	14	12	5	5	27	25	22		
	PrecStd	132	47	43	28	36	28	35	28	34	34	40	51	46		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	13.0	15.7	19.6	25.4	26.5	28.3	29.7	24.6	20.3	15.8	13.7		
		MCWB	10.3	10.7	10.9	12.7	16.1	17.5	19.0	18.6	17.7	15.8	13.6	11.9		
	2%	DB	11.9	11.8	14.1	17.1	23.0	23.5	26.1	26.4	21.9	18.1	14.7	13.0		
		MCWB	10.3	9.6	10.0	11.3	15.2	16.5	18.1	19.0	16.3	14.6	13.1	11.7		
	5%	DB	11.2	11.1	12.8	15.3	20.8	21.7	24.3	24.3	20.5	16.9	13.9	12.3		
		MCWB	9.9	9.1	9.3	10.8	14.4	15.4	17.3	18.0	15.6	14.1	12.4	10.9		
	10%	DB	10.6	10.4	11.8	13.8	18.7	20.1	22.6	22.7	19.2	16.1	13.1	11.4		
		MCWB	9.3	8.5	8.9	9.7	13.5	14.5	16.7	17.3	14.9	13.6	11.6	10.0		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	11.5	12.0	13.9	17.7	18.7	20.6	20.9	18.5	16.6	14.2	12.6		
		MCDB	11.8	12.5	14.6	17.7	23.1	25.3	25.9	26.3	22.5	18.9	14.9	13.1		
	2%	WB	10.7	10.4	11.1	12.2	16.1	17.1	19.1	19.7	17.6	15.6	13.6	11.9		
		MCDB	11.4	11.3	12.9	15.6	21.2	22.0	24.1	24.3	20.7	17.2	14.4	12.7		
	5%	WB	10.2	9.7	10.5	11.3	15.0	16.1	18.2	18.9	16.7	14.9	12.7	11.2		
		MCDB	11.1	10.7	12.0	14.0	19.8	20.2	22.5	23.0	19.2	16.3	13.7	12.1		
	10%	WB	9.4	9.0	9.8	10.5	13.9	15.3	17.4	18.1	15.9	14.1	11.9	10.2		
		MCDB	10.5	10.1	11.2	12.9	17.9	18.9	21.3	21.8	18.2	15.8	12.9	11.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	5.1	6.0	7.0	8.0	7.5	7.9	8.0	7.2	6.3	5.4	4.7		
		MCDBR	4.6	5.1	7.2	9.1	11.5	10.3	11.1	10.6	9.1	6.6	4.7	4.3		
	5% WB	MCWBR	4.4	4.3	4.5	5.0	5.9	5.2	5.2	5.2	5.0	4.5	3.9	4.3		
		MCWBR	4.8	4.9	5.8	7.8	10.2	9.2	9.0	9.0	7.9	5.8	4.5	4.3		
(40) Clear-Sky Solar Irradiance	taub		0.332	0.347	0.384	0.405	0.408	0.404	0.397	0.390	0.376	0.361	0.333	0.322		
		taud	2.360	2.367	2.260	2.226	2.260	2.300	2.326	2.366	2.394	2.414	2.406	2.346		
	Ebn at Noon		681	776	806	833	848	853	853	841	811	749	675	627		
		Edh at Noon	65	83	110	127	128	124	119	109	96	78	62	57		
(44) All-Sky Solar Radiation	RadAvg		0.83	1.53	2.71	4.27	5.15	5.70	5.41	4.52	3.36	1.91	1.00	0.65		
	RadStd		0.06	0.19	0.35	0.42	0.31	0.54	0.45	0.35	0.26	0.16	0.11	0.08		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page